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INTRO	About me: I am an researcher and engineer working on the safety and alignment of artificial intelligence systems. I have extensive experience in dynamic collaborations with mathematicians, archaeologists and clinicians, bridging frontier AI mathematical theory to practical AI applications, providing deterministic methods to affect model behavior (e.g., de-bias , stain , lock , edit or attack) with minimal optimization and negligible cost.	
ACADEMICS	King's College London , Department of Mathematics <i>Postdoctoral Research Associate with Ivan Tyukin (Turing AI Fellow).</i>	2022–2025
	University of Leicester , School of Computing and Mathematical Sciences <i>PhD in Computer Science supervised by Ivan Tyukin and Yudong Zhang.</i>	2019–2023
	University of Sydney School of Mathematics and Statistics, and School of Physics <i>BSc in Science with majors in Physics and Applied Mathematics</i>	2016–2018
EXPERIENCE	Industry Collaboration with TG-0 London <i>Development of reservoir computing algorithms for specific hardware applications.</i>	2024
	Graduate Teaching Assistant with School of Computing and Mathematical Sciences <i>Delivery of a range of undergraduate and master's courses. AFHEA accredited.</i>	2019–2021
	Research Assistant Coordinator with Department of Cardiovascular Sciences, UoL <i>Led a group of research assistants in ML-based knowledge discovery of gene expression data.</i>	2019–2020
WORKING ON NOW	Specialist SLMs for Formal Verification Whole pipeline across data, pre/post-training and evaluations. Current iteration: a 0.5B hierarchical mini "world-model" that simulates the Lean prover and is cheaper to train and inference vs LLM, approaching 7B DeepSeek-Prover-V1 performance. A fully formal and verifiable test-bed.	
SELECT PUBLICATIONS	Harnessing non-adversarial robustness in large language models ICML, Seoul 2026 Spotlight Paper (Top 2.2%) <i>LLMs can falter when input prompts have slight text or format differences. This is due to a shift in the model's internal signal — like a scale that's slightly off-balance. The "scale" can be re-adjusted with a small correction.</i> Through analysis, we trace fragility from semantically-neutral perturbations to systematic biases in module outputs. A cheap closed-form debiasing can be applied with no retraining and often no labeled data. Stealth edits for provably fixing or attacking large language models NeurIPS, Vancouver 2024 Huggingface Live Demo SIAM Front Page <i>LLMs sometimes make factual errors or bad responses, which you can surgically patch by tweaking a few neurons' weights, with guarantees that other answers stay untouched. Unfortunately, this also means an attacker can plant a hidden trigger that's nearly impossible to detect.</i> Through in-depth understanding of LLM and its latent spaces, we show that a single quantity, the separability-based intrinsic dimensionality of a model's latent features, provably governs the selectivity of model editing. Staining and locking computer vision models without retraining ICCV, Hawai'i 2025 Streamlit Live Demo UK Patent <i>To protect your trained model, there are ways to "stain" one with a hidden fingerprint that proves ownership, or "lock" one so that a thief who copies it gets near-useless performance. These protections can be installed directly into the model itself without retraining.</i> We add highly selective detector neurons and disruptor mechanisms directly into the weights and structures of a model. Exploiting feature-space concentration, we gain computable worst-case false-positive bounds.	
TECHNICAL SKILLS	Languages: Python, Matlab (Expert). Mathematica (Proficient), Rust, Typescript, SQL (Comfortable). ML & DL Frameworks: Pytorch, Tensorflow, NumPy, Pandas, Scikit-Learn. CV & NLP Stacks: OpenCV, PIL, Nibabel, FSL, vLLM, LangChain, RAGs. Formal Verification: Lean (Comfortable), Coq, ISabelle (Beginner).	

Neuromorphic tuning of feature spaces to overcome the challenge of low-sample high-dimensional data

This work considers the scenario of mesoscopic sample-size, and based on [theoretical foundations of few-shot learning](#), presents a neuromorphic algorithm to improve latent spaces of pre-trained models.

Quasi-orthogonality and intrinsic dimensions as principled measures of learning and generalisation

This work examines the correlations between the accuracies of trained networks and their initial states and provides a new perspective for zero-shot NAS for finding better architectures.

How adversarial attacks can disrupt seemingly stable accurate classifiers

This work demonstrates that for classifiers on high-dimensional data we have simultaneous susceptibility to adversarial attacks and robustness to random perturbations.

Multiple-instance ensemble for construction of deep heterogeneous committees for high-dimensional low-sample-size data

This work introduces a novel class of stacking methods that utilize attention pooling mechanisms for the construction of both ensembles and cascades for low-sample size medical imaging datasets.

More publications can be found in the list below, or on my [Google Scholar Page](#)

GTA PhD Scholarship, University of Leicester

Lambda Research Grant 2025

Associate Fellow of the Higher Education Academy (AFHEA)

Zhou, Q., Aleshina, E., Lovyagin, A., Somov, O., Seleznyov, M., Panchenko, A., Oseledets, I., Tutubalina, E., Tyukin, I.Y.. 2026. [Harnessing non-adversarial robustness in large language models](#). In proceedings of *International Conference of Machine Learning (ICML)*.

Sutton, O.J.*, **Zhou, Q.***, Leete, F., Gorban, A., Tyukin, I.Y. 2025. [Staining and locking computer vision models without retraining](#). In proceedings of *International Conference on Computer Vision (ICCV)*.

Sutton, O.J.*, **Zhou, Q.***, Wang, W., Higham, D.J., Gorban, A.N., Bastounis, A., Tyukin, I.Y. 2024. [Stealth edits for provably fixing or attacking large language models](#). In proceedings of *Conference on Neural Information Processing Systems (NeurIPS)*.

Sutton, O.J., **Zhou, Q.**, Tyukin, I.Y., Gorban, A.N., Bastounis, A. and Higham, D.J., 2024. [How adversarial attacks can disrupt seemingly stable accurate classifiers](#). *Neural Networks*, 180, 106711.

Tyukin, I.Y., Tykina, T., van Helden, D.P., Zheng, Ze., Mirkes, E.M., Sutton, O., **Zhou, Q.**, Gorban, A.N., Allison, P., 2024. [Coping with AI errors with provable guarantees](#) *Information Sciences*, 678, 106711.

Tyukin, I.Y., Tykina, T., van Helden, D.P., Zheng, Ze., Mirkes, E.M., Sutton, O., **Zhou, Q.**, Gorban, A.N., Allison, P. 2024. [Weakly Supervised Learners for Correction of AI Errors with Provable Performance Guarantees](#). In proceedings of *International Joint Conference on Neural Networks*.

Zhu, H., Wang, W., Ulidowski, I., **Zhou, Q.**, Wang, S., Chen, H., Zhang, Y., 2023. [MEEDNets: Medical Image Classification via Ensemble Bio-inspired Evolutionary DenseNets](#). *Knowledge-Based Systems*, 280, 111035.

Zhou, Q., Wang, S., Zhu, H., Zhang, X., & Zhang, Y. (2023). [Multiple-instance ensemble for construction of deep heterogeneous committees for high-dimensional low-sample-size data](#). *Neural Networks*, 167, 380-399.

Sutton, O., **Zhou, Q.**, Gorban, A.N., Tyukin, I.Y., 2023. [Relative intrinsic dimensionality is intrinsic to learning](#). In proceedings of *International Conference on Artificial Neural Networks (ICANN)*.

Bastounis, A., Gorban, A.N., Hansen, A.C., Higham, D.J., Prokhorov, D., Sutton, O., Tyukin, I.Y., **Zhou, Q.**, 2023. [The Boundaries of Verifiable Accuracy, Robustness, and Generalisation in Deep Learning](#). In proceedings of *International Conference on Artificial Neural Networks (ICANN)*.

Zhou, Q., Sutton, O., Zhang, Y.D., Makarov, V.A., Gorban, A.N., Tyukin, I.Y., 2023. [Neuromorphic tuning of feature spaces to overcome the challenge of low-sample high-dimensional data](#). In proceedings with oral presentation in *International Joint Conference on Neural Networks*.

Zhou, Q., Gorban, A.N., Mirkes, E.M., Bac, J., Zinovyev, A., Tyukin, I.Y., 2022. [Quasi-orthogonality and intrinsic dimensions as measures of learning and generalisation](#). In proceedings with oral presentation in *International Joint Conference on Neural Networks*.

*Equal contributions, alphabetical order

- Zhou, Q.**, Wang, S., Zhang, X., Zhang Y.D., 2021. [WVALE: Weak Variational Autoencoder for Localisation and Enhancement of COVID-19 Lung Infections](#). *Computer Methods and Programs in Biomedicine*, 221, 106883. + Physicsworld [Front Page Article](#).
- Wang, S., Staphathy, S.C., **Zhou, Q.**, Zhang, X., Zhang, Y., 2022. [Secondary pulmonary tuberculosis identification via pseudo-Zernike moment and deep stacked sparse autoencoders](#). *Journal of Grid Computing*, 20, 1-16.
- Yu, X., **Zhou, Q.**, Wang, S. and Zhang, Y.D., 2021. [A systematic survey of deep learning in breast cancer](#). *International Journal of Intelligent Systems*, 37(1), 152-216.
- Zhou, Q.**, Zhu, H., Zhang, X. and Zhang, Y., 2021, December. [Knowledge distillation for secondary pulmonary tuberculosis classification ensemble](#). In Proceedings of the *14th IEEE/ACM International Conference on Utility and Cloud Computing Companion*.
- Wang, S., Celebi, M.E., Zhang, Y.D., Yu, X., Lu, S., Yao, X., **Zhou, Q.**, Miguel, M.G., Tian, Y., Gorriz, J.M. and Tyukin, I., 2021. [Advances in Data Preprocessing for Biomedical Data Fusion: An Overview of the Methods, Challenges, and Prospects](#). *Information Fusion*, 76, 376-421.
- Zhou, Q.**, Zhang, X. and Zhang, Y.D., 2021. [Ensemble learning with attention-based multiple instance pooling for classification of SPT](#). *IEEE Transactions on Circuits and Systems II*, 69(3), 1927-1931.
- Tyukin, I.T., Gorban, A.N., Alkhudaydi, M.H., and **Zhou, Q.**, 2021. [Demystification of Few-shot and One-shot Learning](#). In Proceedings of the *International Joint Conference on Neural Networks*.
- Wang, S.H., Wu, K., Chu, T., Fernandes, S.L., **Zhou, Q.**, Zhang, Y.D. and Sun, J., 2021. [SOSPCNN: Structurally Optimized Stochastic Pooling Convolutional Neural Network for Tetralogy of Fallot Recognition](#). *Wireless Communications and Mobile Computing*, 2021(1), 5792975.
- Chen, X., **Zhou, Q.**, Lan, R., Wang, S.H., Zhang, Y.D. and Luo, X., 2021. [Sensorineural hearing loss classification via deep-HLNet and few-shot learning](#). *Multimedia Tools and Applications*, 80(2), 2109-2122.
- Zhu, M., Zhao, D., Wang, Y., **Zhou, Q.**, Wang, S., Mo, X., Yang, M. and Sun, Y., 2020. [Multi-Slice Radiomic Analysis of Apparent Diffusion Coefficient Metrics Improves Evaluation of Brain Alterations in Neonates With Congenital Heart Diseases](#). *Frontiers in neurology*, 11, 586518.
- Zhang, Y.D., Dong, Z., Wang, S.H., Yu, X., Yao, X., **Zhou, Q.**, Hu, H., Li, M., Jiménez-Mesa, C., Ramirez, J. and Martinez, F.J., 2020. [Advances in multimodal data fusion in neuroimaging: Overview, challenges, and novel orientation](#). *Information Fusion*, 64, 149-187.
- Kang, C., Li, Y., Novak, D., Zhang, Y., **Zhou, Q.** and Hu, Y., 2020. [Brain Networks of Maintenance, Inhibition and Disinhibition During Working Memory](#). *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 28(7), 1518-1527.
- Zhou, Q.**, Mirkes, EM., Zhang, Y., Tyukin, I., Gorban, AN., Yu, X., Wang S., Charlton M., Coats, T., Sims MR., Thompson, JP. 2020, Machine learning technique for severity classification in sepsis patients. *Intensive Care Medicine Experimental 2020*, 8(Suppl 2):73 ID 000473.
- Zhou, Q.**, Zhu, H., Yao, X., Zhang, Y., Murphy, G., Adebayo, A., Wozniak, M., 2020. Obesity-driven differences in gene expression – a machine learning approach. Abstract at *SCTS 2020 conference*.

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